



SERANGOON JUNIOR COLLEGE
General Certificate of Education Advanced Level
Higher 2

Candidate Name

Class

CHEMISTRY

JC2 Preliminary Examination

Paper 1 Multiple Choice

Additional Materials: Data Booklet
 Optical Mark Sheet (OMS)

9729/01

20 September 2018

1 hour

READ THESE INSTRUCTIONS FIRST

On the separate multiple choice OMS given, write your name, subject title and class in the spaces provided.

Shade correctly your FIN/NRIC number.

There are **30** questions in this paper. Answer **all** questions.

For each question there are four possible answers **A, B, C** and **D**.

Choose the one you consider correct and record your choice using a **soft pencil** on the separate OMS.

Each correct answer will score one mark.

A mark will not be deducted for a wrong answer.

You are advised to fill in the OMS as you go along; no additional time will be given for the transfer of answers once the examination has ended.

Any rough working should be done in this question paper.

This document consists of **15** printed pages and **1** blank page.

Answer all questions

- 1** One mole of sulfuric acid is used to make an aqueous solution. The solution contains H_2SO_4 molecules, H^+ ions, SO_4^{2-} ions and HSO_4^- ions.

Which statements are correct?

- (1) The solution contains 6.02×10^{23} sulfur atoms.
- (2) The solution contains an exactly equal number of H^+ ions and HSO_4^- ions.
- (3) One mole of SO_4^{2-} ions contains two moles of electrons

- A** 1 only
- B** 1 and 2 only
- C** 2 and 3 only
- D** 1 and 3 only

- 2** The table refers to the electron distribution in the second shell of an atom with eight protons. Which row is correct for this atom?

	Orbital shape 		Orbital shape 	
	Orbital type	Number of electrons	Orbital type	Number of electrons
A	p	2	s	4
B	p	4	s	2
C	s	2	p	4
D	s	4	p	2

- 3 50 cm³ of a 0.10 mol dm⁻³ solution of a metallic salt was found to react exactly with 25.0 cm³ of 0.10 mol dm⁻³ aqueous sodium sulfite. In this reaction, the sulfite ion is oxidised as follows:



What is the new oxidation number of the metal in the salt if its original oxidation number was +3?

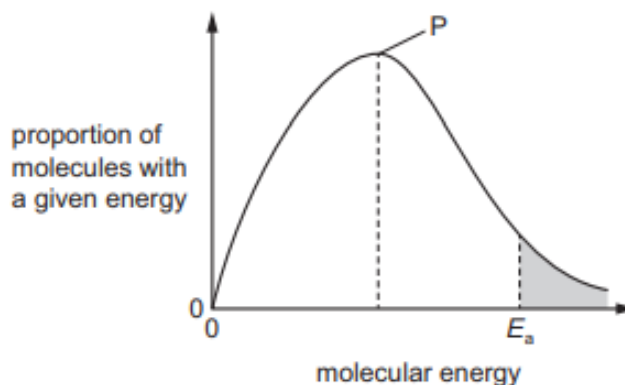
- A +1
B +2
C +4
D +5
- 4 A 2 g sample of hydrogen at temperature T and of volume V exerts a pressure p .

Deuterium, ${}^2_1\text{H}$ is an isotope of hydrogen.

Which of the following would also exert a pressure of p at the same temperature T?

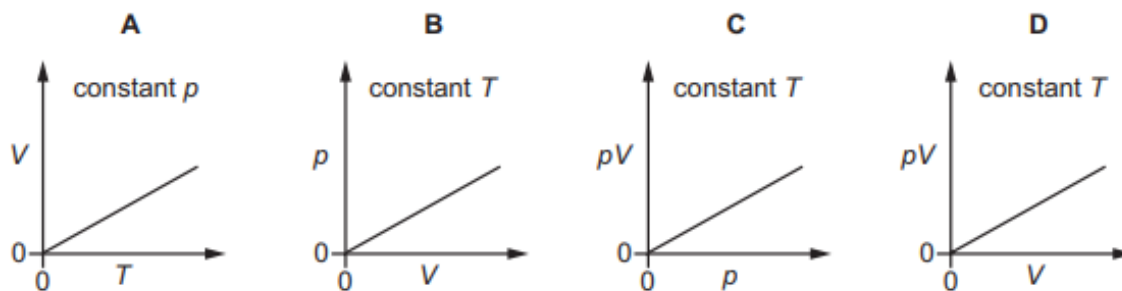
- A A mixture of 2 g of hydrogen and 2 g of deuterium of total volume 2V
B A mixture of 1 g of hydrogen and 2 g of deuterium of total volume 2V
C A mixture of 1 g of hydrogen and 2 g of deuterium of total volume V
D A mixture of 1 g of hydrogen and 1 g of deuterium of total volume V

- 5 The diagram shows the Boltzmann distribution of energies in a gas. The gas can take part in a reaction with an activation energy, E_a .

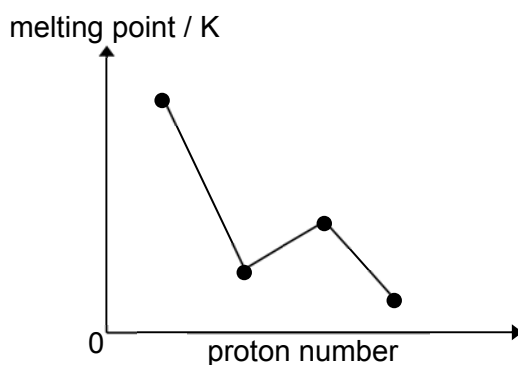


Which statement is correct?

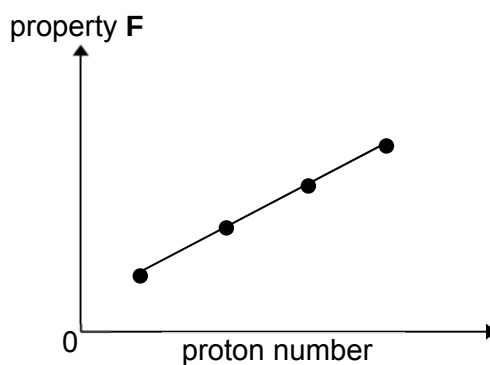
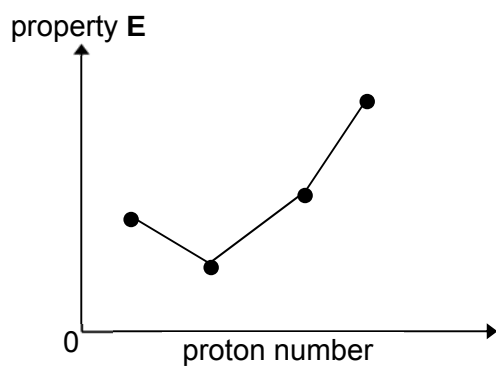
- A If temperature is increased, peak P will be lower and E_a will move to the right.
 - B If temperature is increased, peak P will be higher and E_a will not move
 - C If temperature is decreased, peak P will be the same and E_a will move to the left.
 - D If temperature is decreased, peak P will be higher and E_a will not move.
- 6 Which diagram correctly describes the behaviour of a fixed mass of an ideal gas? (T is measured in K.)



- 7 The diagram represents the melting points of four consecutive elements in the third period of the Periodic Table.



The sketches below represent another two properties of the elements.



What are properties **E** and **F**?

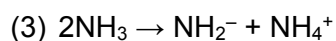
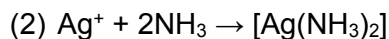
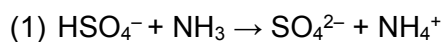
- | property E | property F |
|--------------------------------------|-------------------|
| A third ionisation energy | electronegativity |
| B number of valence electrons | boiling point |
| C ionic radius | nuclear charge |
| D electrical conductivity | atomic radius |

- 8 When 60 cm³ of 0.1 mol dm⁻³ of sulfuric acid and 40 cm³ of 0.2 mol dm⁻³ sodium hydroxide were mixed in a styrofoam cup, the temperature rose by 6.5 °C.

Calculate the standard enthalpy change of neutralisation. Assume that the specific heat capacity of the solution is 4.2 Jg⁻¹K⁻¹.

- A + 34.1 kJ mol⁻¹
 B + 45.5 kJ mol⁻¹
 C - 34.1 kJ mol⁻¹
 D - 45.5 kJ mol⁻¹

- 9 In which reactions does NH₃ behave as a Brønsted-Lowry acid?



- A 1 and 2 only
 B 1 and 3 only
 C 1 only
 D 3 only

- 10 A current of 0.2 ampere passing for 5 hours through a solution of gold ions deposits a mass of 2.45 g of gold on the cathode. Which of these expressions gives the charge on a gold ion?

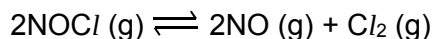
A
$$\frac{2.45 \times 0.2 \times 5 \times 60 \times 60}{197 \times 96500}$$

B
$$\frac{0.2 \times 5 \times 60 \times 60 \times 197}{96500 \times 2.45}$$

C
$$\frac{2.45 \times 96500}{197 \times 0.2 \times 5 \times 60 \times 60}$$

D
$$\frac{197 \times 0.2 \times 5 \times 60 \times 96500}{2.45}$$

- 11** Pure nitrosyl chloride, NOCl gas, was heated at 320°C in a 2.0 dm^3 vessel. At equilibrium, 30% of the NOCl gas had dissociated according to the equation below and the total pressure was $p\text{ atm}$.



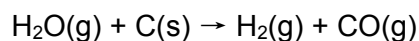
What is value of the equilibrium constant, K_p ?

- A** $\frac{17.9}{p}$ **B** $\frac{41.7}{p}$ **C** $0.0120p$ **D** $0.0130p$

- 12** Calculate the resultant pH of the solution when 10 cm^3 of hydrochloric acid with a concentration of 0.015 mol dm^{-3} was added to a 25 cm^3 sample of ammonia with a concentration of 0.25 mol dm^{-3} .

(K_b of ammonia = $1.778 \times 10^{-5}\text{ mol dm}^{-3}$)

- A** 10.9
B 8.25
C 7.64
D 9.25
- 13** Hydrogen can be made from steam according to the following equation:



The Gibbs free energy change of reaction at two different temperature are shown

$\Delta G_1 = +78\text{ kJ mol}^{-1}$ at 378 K

$\Delta G_2 = -58\text{ kJ mol}^{-1}$ at 1300 K

Which row of the table gives the correct sign of ΔH and ΔS for this reaction?

	ΔH	ΔS
A	–	–
B	–	+
C	+	–
D	+	+

- 14 An experiment was carried out to investigate the initial rate of reaction between potassium peroxodisulphate, $K_2S_2O_8$, an oxidising agent, and potassium iodide, KI.

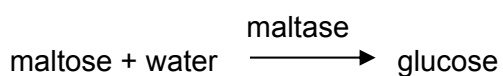
The initial volumes of the $K_2S_2O_8$ and KI solutions in the mixture together with the time taken for the mixture to darken for the various experimental runs are given below.

Volume of $K_2S_2O_8$ / cm^3	Volume KI / cm^3	Volume of water / cm^3	time taken to darken / s
10	20	10	35
5	20	15	70
10	8	22	88
20	40	20	y

Select the correct option for the following reaction.

	Order with respect to $K_2S_2O_8$	Order with respect to KI	y / s
A	1	1	70.0
B	1	2	17.5
C	1	1	35.0
D	2	1	17.5

- 15 The enzyme maltase speeds up the reaction between maltose and water.

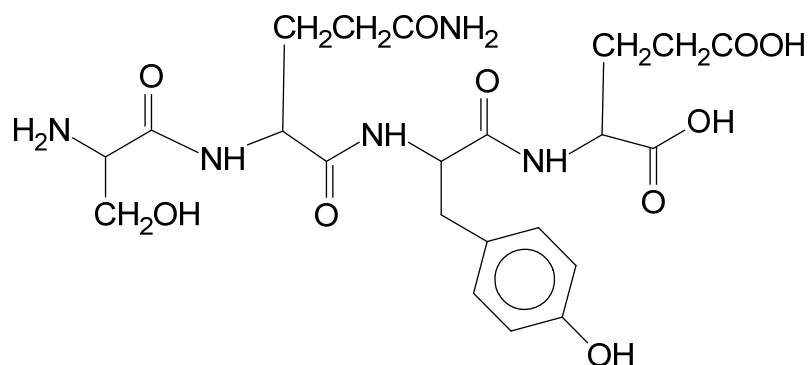


Maltase shows **specificity**.

Which statement describes the **specificity** of maltase?

- A** Maltase is a biological catalyst and it is a type of protein.
- B** Maltase is most effective between pH 6.1 and pH 6.8.
- C** Maltase lowers the activation energies of the reactions it catalyses.
- D** Maltase only speeds up a small number of chemical reactions.

16 The diagram shows the structure of the tetrapeptide, **J**.

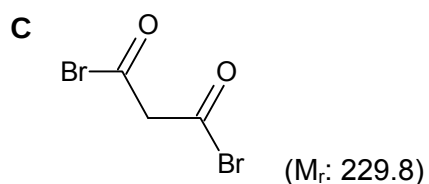
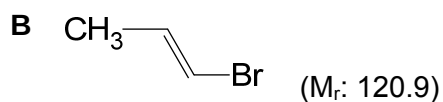
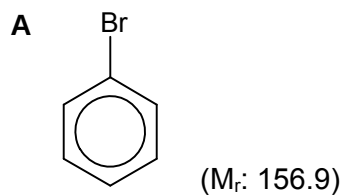


Which statements are correct?

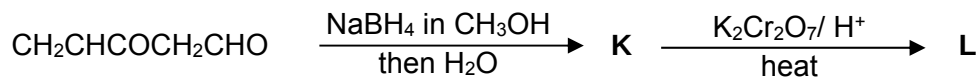
- (1) When 1 mol of **J** reacts with hot NaOH(aq) until no further reaction occurs, 8 mol of NaOH will react.
- (2) When 1 mol of **J** reacts with hot HCl(aq) until no further reaction occurs, 5 mol of HCl will react.
- (3) When 1 mol of **J** reacts with ethanoyl chloride, 3 mol of ethanoyl chloride will react forming ester or amide.
- (4) When 1 mol of **J** reacts with Na(s), 4 mol of hydrogen gas will be given out.

- A** 1 and 2 only
B 2 and 3 only
C 1, 2 and 4 only
D 3 and 4 only

- 17 Which of these will produce the most silver bromide precipitate when a 1 g sample reacts with excess hot sodium hydroxide, followed by silver nitrate solution?



- 18 Identify the final product L in this sequence of reactions.



- A CH₂CHCOCH₂COOH
 B CH₃CH₂COCH₂COOH
 C CH₂(OH)CH(OH)COCH₂CH₂OH
 D CH₂(OH)CH(OH)CH(OH)CH₂CH₂OH

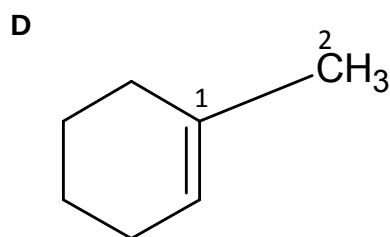
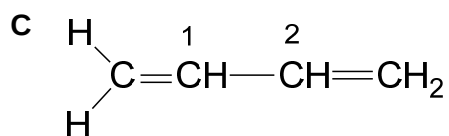
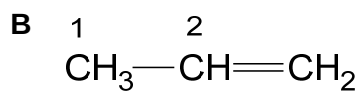
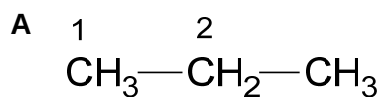
19 An alcohol **M** with molecular formula $C_4H_{10}O$ is oxidised by acidified potassium dichromate(VI) under certain conditions to give **N**.

- **N** does not produce a yellow precipitate with aqueous alkaline iodine
- **N** gives a reddish brown precipitate when reacted with Fehling's solution

How many isomers of alcohol **M** could result in the observations for **N**?

- A** 1
B 2
C 3
D 4

20 Which of the following compounds has the shortest C1-C2 bond length?

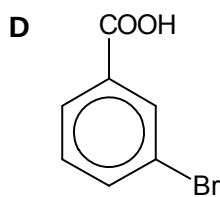
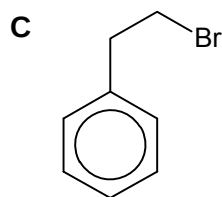
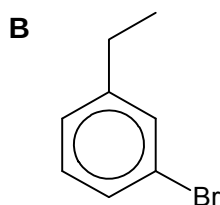
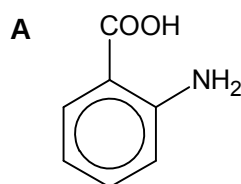


- 21 Maleic acid is used in the food industry and for stabilising drugs. It is the cis-isomer of butenedioic acid and has the structural formula $\text{HO}_2\text{CCH}=\text{CHCO}_2\text{H}$.

What is the product formed from the reaction of maleic acid with cold, dilute, acidified manganate(VII) ions?

- A $\text{HO}_2\text{CCH}(\text{OH})\text{CH}(\text{OH})\text{CO}_2\text{H}$
 B $\text{HO}_2\text{CCO}_2\text{H}$
 C $\text{HO}_2\text{CCH}_2\text{CH}(\text{OH})\text{CO}_2\text{H}$
 D $\text{HO}_2\text{CCOCOCO}_2\text{H}$

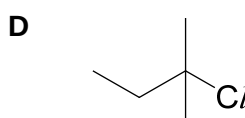
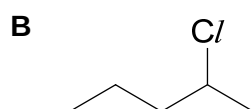
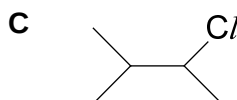
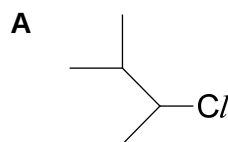
- 22 Which one of the following compounds **cannot** be synthesised from ethylbenzene?



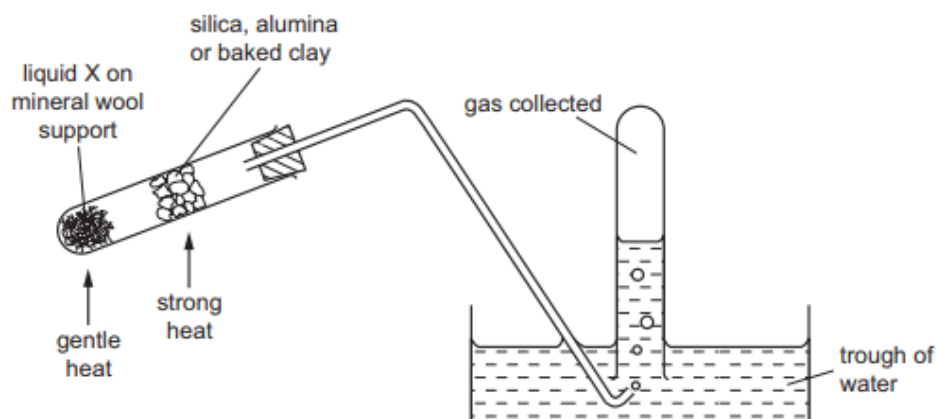
- 23 Structural isomerism and stereoisomerism should be considered when answering this question.

A colourless liquid, $\text{C}_5\text{H}_{11}\text{Cl}$, exists as a mixture of two optical isomers.

When heated with sodium hydroxide in ethanol, a mixture of only **two** alkenes is formed. What could the colourless liquid be?



- 24 The diagram shows an experimental set-up which can be used in several different experiments.

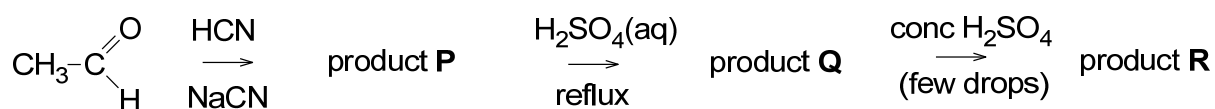


Which processes could be demonstrated by using the above apparatus?

- (1) oxidation of ethanol (liquid X)
- (2) dehydration of ethanol (liquid X)
- (3) cracking of paraffin (liquid X)

- A 1, 2 and 3
 B 1 and 2 only
 C 2 and 3 only
 D 3 only

- 25 Ethanal, CH_3CHO , is used to make product **R** in a three-stage synthesis.



Two molecules of **Q** react to give one molecule of **R** plus two molecules of water.

R does not react with sodium.

What is the molecular formula and empirical formula of **R**?

	Molecular formula	Empirical formula
A	$\text{C}_3\text{H}_4\text{O}_2$	$\text{C}_3\text{H}_4\text{O}_2$
B	$\text{C}_6\text{H}_8\text{O}_4$	$\text{C}_3\text{H}_4\text{O}_2$
C	$\text{C}_3\text{H}_5\text{O}_2$	$\text{C}_3\text{H}_5\text{O}_2$
D	$\text{C}_6\text{H}_{10}\text{O}_5$	$\text{C}_6\text{H}_{10}\text{O}_5$

- 26** Chlorofluoroalkanes have been used as the refrigerant in refrigerators but care has to be taken in disposing of old refrigerators.

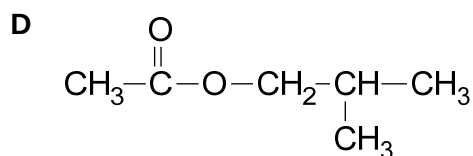
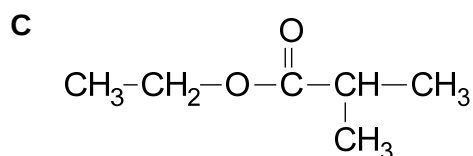
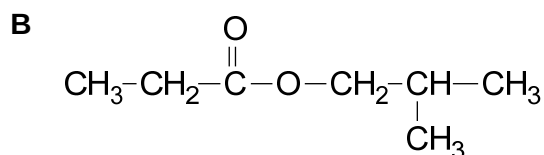
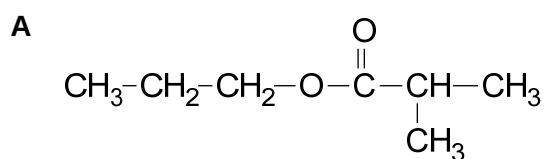
Which statements about chlorofluoroalkanes are correct?

- (1) C–Cl bonds more readily undergo homolytic fission than C–F bonds.
- (2) Care is taken in the disposal of old refrigerators because of possible ozone depletion.
- (3) $\text{C}_2\text{H}_4\text{ClF}$ is more volatile than C_2H_6 .

- A** 2 only
- B** 1 and 2 only
- C** 2 and 3 only
- D** 1, 2 and 3

- 27** Ethyl propanoate is refluxed with aqueous sodium hydroxide. The alcohol produced is then reacted with methyl propanoic acid to make a second ester.

What is the structural formula of this second ester?



- 28 Ethanedioic acid has the formula $\text{HO}_2\text{CCO}_2\text{H}$.

What is the formula of aluminium ethanedioate?

- A AlC_2O_4
- B $\text{Al}(\text{C}_2\text{O}_4)_3$
- C $\text{Al}_2\text{C}_2\text{O}_4$
- D $\text{Al}_2(\text{C}_2\text{O}_4)_3$

- 29 Which of the following processes lead to an increase in entropy?

- (1) Diffusion of air fresher in the lecture theatre.
- (2) Combustion of a piece of charcoal to form $\text{CO}_2(\text{g})$ and $\text{H}_2\text{O}(\text{g})$.
- (3) Desalination of sea water by reverse osmosis (solvent passes from a dilute solution to a concentrated solution).

- A 1 only
- B 1 and 2 only
- C 2 and 3 only
- D 1, 2 and 3

- 30 How many structural isomers with the molecular formula $\text{C}_5\text{H}_{10}\text{O}_2$ give infra-red absorptions both at approximately 1300 cm^{-1} and at approximately 1740 cm^{-1} ?

- A 3
- B 5
- C 7
- D 9

END OF PAPER 1

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